

Explanation of AWS service:

1. EC2 (Elastic Compute Cloud)

- **Type:** Compute
 - **Description:** EC2 provides scalable computing power in the cloud. It allows users to launch virtual servers (instances) with different configurations based on their requirements.
 - **Use Cases:** Hosting websites, running applications, processing large datasets, machine learning model training.
 - **Key Features:** Auto Scaling, Elastic Load Balancing, Various instance types (On-demand, Reserved, Spot, Dedicated).
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2. EFS (Elastic File System)

- **Type:** Storage
 - **Description:** EFS is a fully managed NFS (Network File System) that provides scalable file storage for EC2 instances. It automatically grows and shrinks as you add or remove files.
 - **Use Cases:** File sharing across multiple EC2 instances, media processing, big data analytics.
 - **Key Features:** Automatic scaling, shared access, high availability, POSIX compatibility.
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3. EBS (Elastic Block Store)

- **Type:** Storage
 - **Description:** EBS provides persistent block storage for EC2 instances. It behaves like a hard drive and is attached to EC2 instances for storing data.
 - **Use Cases:** Databases, log storage, backup and recovery.
 - **Key Features:** SSD & HDD options, snapshot backups, encryption, high durability.
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4. S3 (Simple Storage Service)

- **Type:** Object Storage

- **Description:** S3 is an object storage service that allows users to store and retrieve unlimited amounts of data. It provides high durability and availability.
 - **Use Cases:** Backup storage, hosting static websites, storing media files, big data analytics.
 - **Key Features:** Versioning, lifecycle policies, encryption, access control, event notifications.
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5. Route 53

- **Type:** Networking & DNS
 - **Description:** Route 53 is a highly scalable domain name system (DNS) web service that translates domain names into IP addresses.
 - **Use Cases:** Domain registration, traffic routing, failover configurations, latency-based routing.
 - **Key Features:** Global DNS management, health checks, traffic routing, domain registration.
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6. RDS (Relational Database Service)

- **Type:** Database
 - **Description:** RDS is a managed database service that supports multiple relational databases such as MySQL, PostgreSQL, MariaDB, Oracle, and SQL Server.
 - **Use Cases:** Running relational databases, automatic backups, failover management.
 - **Key Features:** Automated backups, Multi-AZ deployment, high availability, read replicas.
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7. Lambda

- **Type:** Compute (Serverless)
- **Description:** AWS Lambda is a serverless computing service that runs code in response to events. You only pay for the compute time used.
- **Use Cases:** Event-driven applications, automated workflows, API backends, IoT applications.

- **Key Features:** No server management, automatic scaling, pay-per-execution, supports multiple programming languages.
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8. SageMaker

- **Type:** Machine Learning
 - **Description:** SageMaker is a fully managed machine learning (ML) service that enables developers and data scientists to build, train, and deploy ML models.
 - **Use Cases:** Image recognition, recommendation systems, predictive analytics.
 - **Key Features:** Pre-built algorithms, Jupyter notebooks, auto-scaling, distributed training.
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9. ECS (Elastic Container Service)

- **Type:** Containers
 - **Description:** ECS is a fully managed container orchestration service that allows you to run and scale containerized applications using Docker.
 - **Use Cases:** Running microservices, deploying containerized applications.
 - **Key Features:** Integration with Fargate (serverless), auto-scaling, networking flexibility.
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10. Cloud9

- **Type:** Development
 - **Description:** AWS Cloud9 is a cloud-based IDE (Integrated Development Environment) for writing, running, and debugging code in the cloud.
 - **Use Cases:** Cloud-based coding, collaborative development, debugging.
 - **Key Features:** Browser-based editor, supports multiple programming languages, preconfigured AWS SDKs.
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11. WAF (Web Application Firewall)

- **Type:** Security

- **Description:** WAF helps protect web applications from common threats such as SQL injection and cross-site scripting (XSS).
 - **Use Cases:** Protecting websites and APIs, blocking malicious traffic.
 - **Key Features:** Custom rules, bot mitigation, OWASP Top 10 protection.
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12. IAM (Identity and Access Management)

- **Type:** Security
 - **Description:** IAM is a security service that manages users, groups, and permissions in AWS. It enables fine-grained access control.
 - **Use Cases:** User authentication, role-based access control, managing AWS credentials.
 - **Key Features:** Multi-factor authentication (MFA), policies, role-based permissions.
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13. Kinesis

- **Type:** Data Streaming
 - **Description:** Kinesis enables real-time data streaming and analytics for collecting, processing, and analyzing data.
 - **Use Cases:** Real-time analytics, IoT data processing, log monitoring.
 - **Key Features:** Real-time processing, high scalability, integrates with AI/ML services.
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14. DynamoDB

- **Type:** NoSQL Database
 - **Description:** DynamoDB is a fully managed NoSQL database service designed for high-speed performance and scalability.
 - **Use Cases:** Gaming leaderboards, IoT applications, session storage.
 - **Key Features:** Serverless, auto-scaling, multi-region replication, fast performance.
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15. SNS (Simple Notification Service)

- **Type:** Messaging

- **Description:** SNS is a fully managed messaging service that enables applications to send notifications via SMS, email, or push notifications.
 - **Use Cases:** Sending alerts, broadcasting messages, IoT notifications.
 - **Key Features:** Pub/Sub messaging model, supports multiple endpoints, integrates with Lambda & SQS.
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16. Rekognition

- **Type:** AI/ML (Image & Video Analysis)
 - **Description:** Rekognition is an AI-powered image and video analysis service that can detect faces, objects, and text in media.
 - **Use Cases:** Facial recognition, content moderation, automatic image tagging.
 - **Key Features:** Face detection, sentiment analysis, object recognition, celebrity detection.
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Summary of Key AWS Services

Service	Type	Description
EC2	Compute	Virtual servers in the cloud
EFS	Storage	Scalable shared file storage
EBS	Storage	Block storage for EC2
S3	Storage	Object storage service
Route 53	Networking	DNS and domain name management
RDS	Database	Managed relational database service
Lambda	Compute	Serverless function execution
SageMaker	ML	Machine learning model training and deployment
ECS	Containers	Container orchestration service
Cloud9	Development	Cloud-based IDE for coding
WAF	Security	Web application firewall
IAM	Security	User and permission management

Kinesis	Data Streaming	Real-time data processing
DynamoDB	Database	NoSQL database
SNS	Messaging	Notification and alert service
Rekognition	AI/ML	Image and video analysis

MCQS into One Word Answer

1. **Cloud computing provides which type of resources to customers?**
Answer: Virtualized
2. **Cloud computing is based on the concept of?**
Answer: Virtualization
3. **Which of these is not a cloud computing service model?**
Answer: DBaaS
4. **Which cloud model offers computing resources over the internet?**
Answer: Public cloud
5. **Which cloud model is dedicated to a single organization?**
Answer: Private cloud
6. **A combination of both public and private cloud is called?**
Answer: Hybrid cloud
7. **Cloud computing service models are also known as?**
Answer: SPI models
8. **Which service model provides virtualized computing resources over the internet?**
Answer: IaaS
9. **Which service model provides a runtime environment for application development?**
Answer: PaaS
10. **Which service model provides ready-to-use software applications?**
Answer: SaaS

11. **Which of the following is an example of SaaS?**
Answer: Gmail
12. **Which of the following is an example of PaaS?**
Answer: Google App Engine
13. **Which of the following is an example of IaaS?**
Answer: AWS EC2
14. **Which cloud model is best suited for government organizations?**
Answer: Community cloud
15. **The on-demand availability of computing resources over the internet is known as?**
Answer: Cloud computing
16. **Which among the following is a characteristic of cloud computing?**
Answer: All of these (On-demand self-service, Broad network access, Resource pooling)
17. **What does multi-tenancy mean in cloud computing?**
Answer: Multiple customers share the same resources
18. **Which technology allows multiple operating systems to run on a single hardware platform?**
Answer: Virtualization
19. **Cloud computing enables which type of scaling?**
Answer: Both (Vertical and Horizontal)
20. **Which of the following is a security concern in cloud computing?**
Answer: Data privacy
21. **Which service model provides the highest level of control to users?**
Answer: IaaS
22. **Which service model requires the least maintenance from users?**
Answer: SaaS
23. **Which of the following is a benefit of cloud computing?**
Answer: All of these (Cost efficiency, Scalability, Business continuity)
24. **What does pay-as-you-go pricing in cloud computing mean?**
Answer: Users pay only for what they use
25. **Which type of virtualization is used for running multiple operating systems on a single machine?**
Answer: Server virtualization

26. **Which cloud deployment model provides exclusive cloud access to a single organization?**
Answer: Private cloud
27. **Which type of cloud offers services to multiple customers over the internet?**
Answer: Public cloud
28. **Which type of cloud is a combination of public and private clouds?**
Answer: Hybrid cloud
29. **Which of these is an advantage of using cloud computing?**
Answer: Scalability
30. **Which hypervisor type runs directly on the hardware?**
Answer: Type 1
31. **Which hypervisor type runs over the host operating system?**
Answer: Type 2
32. **Which of these is a risk in cloud computing?**
Answer: Data security
33. **Which of these is NOT a cloud computing characteristic?**
Answer: Manual resource allocation
34. **Which of the following cloud models is best suited for startup companies?**
Answer: Public cloud
35. **Which of the following ensures security in cloud computing?**
Answer: Encryption
36. **Which of the following best defines cloud elasticity?**
Answer: Ability to scale resources dynamically
37. **What is the main function of a hypervisor?**
Answer: Manage virtual machines
38. **Which cloud model is best for government organizations and universities?**
Answer: Community cloud
39. **Which of the following is NOT a key component of cloud computing?**
Answer: Mainframe computing
40. **Which cloud service model is best for developers to build applications without managing infrastructure?**
Answer: PaaS

41. Which type of cloud model allows organizations to combine on-premise infrastructure with cloud services?
Answer: Hybrid cloud
42. Which of the following is a major challenge in cloud computing?
Answer: Data security
43. Which cloud characteristic allows users to access services over the internet from anywhere?
Answer: Broad network access
44. Which feature of cloud computing allows multiple users to share the same resources efficiently?
Answer: Multi-tenancy
45. Which cloud computing model is best for small businesses with limited IT resources?
Answer: Public cloud
46. Which of the following best describes "Infrastructure as a Service (IaaS)"?
Answer: Virtualized computing resources over the internet
47. Which cloud service model provides a ready-made platform for application development?
Answer: PaaS
48. Which of the following is NOT an advantage of cloud computing?
Answer: High initial investment
49. Which of the following helps ensure cloud service availability?
Answer: Load balancing
50. Which cloud computing feature allows businesses to increase or decrease resources based on demand?
Answer: Elasticity
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The second list of one-word answer MCQs:

1. Which of these attributes in a system could be brought by computing infrastructure virtualization?
Answer: All of these (Security, Dynamic behavior, Flexibility)
2. The physical system on which virtual machines run is called?
Answer: Host machine

3. **The software layer implementing virtualization is known as?**
Answer: Hypervisor
4. **In server virtualization, the relationship between the host machine and guest machine is?**
Answer: One-to-many
5. **Is it possible to install an operating system in a virtual machine?**
Answer: Yes
6. **Type 1 hypervisors are installed over?**
Answer: Physical resource
7. **Type 2 hypervisors are installed over?**
Answer: Host operating system
8. **Operating systems and applications made for one computer architecture can be installed on some other architecture in?**
Answer: Emulation-based server virtualization
9. **Guest operating systems remain unaware that they are running in a virtual environment in?**
Answer: Full virtualization
10. **Hypervisor-dependent (modified) versions of guest operating systems are required in?**
Answer: Para-virtualization
11. **Virtualization overhead of hypervisor is maximum in case of?**
Answer: Full virtualization
12. **Virtual machines running over a physical host do not have different operating systems in?**
Answer: Operating system-level virtualization
13. **The most popular open-source hypervisor available in the market is?**
Answer: Xen
14. **Process virtual machines are made to run?**
Answer: Some specific application
15. **Server virtualization creates scope for?**
Answer: Both b & c (Fast installation of servers, Server portability)
16. **Resource abstraction in cloud computing is implemented through?**
Answer: Resource virtualization

17. **Virtual machine monitor (VMM) is the other name of?**

Answer: Hypervisor

18. **In operating system-level virtualization, different OS distributions of the same kernel can be run in virtual servers.**

Answer: True

19. **Cloud computing IaaS services can be developed without implementing resource virtualization also.**

Answer: False

20. **Which among the following must be maintained to ensure the security of virtualization?**

Answer: All of these (Hardening the hypervisor, Restricting physical access to the host machine, Maintaining one function per VM)

21. **The 'single point' in the 'single point of failure problem' of virtualization is?**

Answer: VMM

22. **In 'VM escape', which among the following happens?**

Answer: Accessing host OS by escaping VM

23. **Inactive virtualization machines may cause security threats to virtualization?**

Answer: True

24. **The benefits of infrastructure virtualization include?**

Answer: All of these (Better utilization of resources, Cost saving for computation, Easier capacity expansion)

REVIEW QUESTIONS

How virtualization enables multi-tenancy in computing?

In virtualization, the users as well as the applications can only access virtualized computing resources

(like processor, memory, disk, switch etc.). These virtualized computing resources are made out of

support from physical computing resources. One physical device may empower multiple virtual

devices (of its own type) at same or different point of time which are consumed by users (tenants) not

known to each other. This flexibility is there since applications or users are not tied with any physical

resources, rather virtual resource are being allocated according to the demand of applications, thus

one physical resource can support multiple tenants.

When does it become necessary to implement emulation-based virtualization?

Normal virtualization is simulation-based. Software simulates the actual hardware components into

set of software-based (virtual) components. Architectural structure of these virtual components

remains same as the actual physical components. Hence, software systems (operating system or

applications) fit for the architecture (or instruction set) of the physical resources can be installed over

the corresponding virtual resources.

When situation demands that software made for one architecture (that is written for one type of

instruction set) has to be installed over hardware resources having some other architecture (made

for some other instruction set); then only way is to create the required set of virtual devices through

emulation-based virtualization. In emulation-based virtualization, the architecture of virtual devices

and the underlying physical devices need not to match.

Why is hypervisor also called virtual machine monitor?

Hypervisor produces all of the virtual computing resources. Virtual machines are created over the

layer of hypervisor using these virtual resources. It is the responsibility of the hypervisor to manage

activities, monitor functionalities and provide support to the virtual machines. This is why hypervisor

is also called as virtual machine monitor or VMM.

How virtualization empowers computing using resource pool?

In contrast to the fact that one physical resource can support multiple virtual devices (through multi-tenancy), it is also possible to form one powerful virtual device combining the power of multiple

physical resources. One 6GB memory can be formed combining three 2GB RAMs. Again those three

Resource Virtualization 135

RAMs can be replaced by two 3GB RAMs without disrupting the service or without any knowledge

of applications using them. All of these mapping of virtual resources with underlying physical pool

of resources occur dynamically to support maximum resource utilization. Decoupling of applications

from hardware allows administrators to think more about aggregate computing capacity and less about individual hosts. Thus, the virtualization endorses to gain maximum benefit of pooled

resources.

What is sandboxing?

In computing, a sandbox is a restricted or isolated environment for execution of application. This

environment does not allow to execute all functionalities by prohibiting certain functions for security

purpose. This environment works as virtual lab where un-trusted programming codes or applications

can be tested without allowing it to cause any harm.

Virtualization and sandboxing are not same. Rather, sandboxing can be seen as a special and partial example of virtualization. But virtualization sometimes referred as sandboxing as it creates a

protected enclosure that mimics a real-life computing infrastructure and thus malicious applications

cannot damage the actual resources.

Cloud vendors often use sandboxing for security. For instance, Google App Engine uses sandbox

feature to keep applications isolated to ensure that all App Engine applications run in a restricted

environment. This also ensures the security of the applications against the attacks.

What is the difference between sandboxing and virtualization?

The objective of sandboxing and virtualization are entirely different. Sandboxing is to isolate a

running program or application from the physical computing system so that the program cannot

harm the system. It is used to verify untrusted applications. While virtualization is used to create

entire machine environment, sandboxing software provides minimal functionalities required for

application execution or testing and withheld codes that can harm the server and data.

Resource abstraction through virtualization: What does this mean?

When working in virtualized environment, no one can access physical resources directly. Applications

run upon virtual machines and they can never know which physical devices are actually supporting

those virtual devices or where they are located. Thus, the actual resource components, their architecture, and system configuration remain out of everyone's sight. Virtualization thus adds a layer

of abstraction over physical resources.

When is operating system-level virtualization useful?

Server or machine virtualization facilitates the running of multiple virtual or guest machines on a

single physical or host machine. In hypervisor-based virtualization, the different guest machines

running over a host need not to run same OS as the host. They are free to run any OS and multiple

VMs running over a host may run different OS simultaneously.

System level (or operating system level) virtualization is an option when it is known that all of

the virtual servers will use same Linux-based operating system. In system-level virtualization, virtual

servers share one single OS kernel (that of the host OS) and hence can run same operating system

only.

What is the importance of the security reference model in cloud computing?

Cloud computing is made for collaboration and accessed remotely. It allows de-perimeterization

that is admission of external systems into organization's network security boundary. Users may not

own a major part of the systems they work in. Accountability of service providers plays an important

role in this computing model. Thus, the security erupts as a basic concern. A guideline for managing

security makes the scenario easy for consumers while moving into cloud environment. The model

distinguishes all of the possible formations of cloud and highlights their pros and cons both.

Were the four dimensions of cloud security model relevant in traditional computing too?

Outsourcing of computing was present in traditional computing also. Thus, the sourcing dimension

(third dimension of cube model) stands relevant there. In traditional approach, data center location

could either be on-premises or off-premises. Hence data storage location could be internal or

external. Eventually, the data boundary dimension (first dimension of cube model) is also relevant

there. The second dimension becomes relevant as the ownership of technology used in traditional

computing could be both proprietary and open. Now in relation with the third dimension, it we can

say that the traditional computing used to prefer the perimeterized security approach. But in special

cases, de-perimeterization could be done through virtual private network. Consequently, the cloud

computing is primarily created to take the advantages of de-perimeterized computing approach.

What is Cloud Security Alliance?

Cloud Security Alliance (CSA) is a non-profit organization formed by a group of industry leaders,

focussing on implementation of wide range of initiatives in cloud security. CSA promotes cloud

security awareness programs, regularly publishes security guidance on cloud computing and offers

security certifications. The organization was formed in 2008 and registered in United States.

IT outsourcing is more secured under which model: traditional computing or cloud computing?

Outsourcing model of computing can be of two types as perimeterized and de-perimeterized.

Perimeterization keeps the computing environment under the network boundary of consumer

enterprise. Thus, it is sometimes considered as more secured option than de-perimeterization because

all resources, especially storage and others remain under the control of the enterprise. Otherwise, the

security in outsourcing mainly depends on the reputation and expertise of the computing vendor.

In this regard, public cloud services offered by reputed computing vendors are often considered as

more secured, reliable and robust in comparison to traditional model of outsourcing.

Is there any way of enjoying facilities of cloud computing without compromising security?

This is absolutely possible with private cloud deployment model. In private cloud, consumers can

get all benefits of cloud computing like resource pooling, on-demand service, elasticity etc. Only

thing they do not get is multi-tenancy facility which makes public cloud services more cost-effective.

Private clouds can provide traditional computing like security confidence both for insourced and

outsourced services.

Computing infrastructure as service: What does it mean and how is it possible?

When computing infrastructure becomes available as service that means people need not to acquire

physical computing systems for getting their works done; be it about performing computational

task, forming network of computers or storing data. All kinds of computing infrastructural facilities

become readily available in virtual mode which users can access through a thin client interface

(usually a browser) from any computing device. Users can perform all activities using these virtual

computing resources just like they used to do with actual physical computers. Computing vendors

arrange, manage and deliver the resources to users over internetwork/Internet. This is called as the

computing infrastructure as service.

What activities can be performed by consumers of PaaS facility?

Consumers of PaaS get a computing platform which is actually an application development and

deployment environment. PaaS delivers virtual machines loaded with operating system and

integrated with the runtime(s) of the desired development and/or deployment platform.

Consumers

can develop applications over the delivered platform and debug or test them. They can also install

and run supported applications over a PaaS offering.

How can a private cloud be enabled to expand into hybrid cloud environment?

For a private cloud solution to expand into hybrid cloud environment, it must be integrated with

some public cloud solutions. This is only possible when the private cloud solution is compatible

with the public cloud solution. There are instances where open-source cloud solutions (those are

used to build private cloud) have gone into agreement with commercial public cloud providers

for compatibility. Several open-source cloud solutions like Eucalyptus, OpenNebula and Nimbus

implement Amazon cloud's APIs and thus can integrate with Amazon cloud in order to build hybrid

cloud.

Do all service providers follow NIST model for cloud service development?

Several architectural models have been proposed for cloud computing over the years.

Among

them, the model proposed by NIST has been recognized as the globally accepted cloud reference

architecture. Service providers generally do not disclose their cloud computing models. But major

cloud computing vendors have their own cloud architectural model and it is most likely that they

follow their own model.

What is the difference between elasticity and scalability in cloud computing?

One important feature of cloud computing is its scalable nature. A scalable architecture handles

growing workload by increasing resource capacity in same proportion. Thus scalability of cloud

architecture helps in delivering resources as per demand of application.

Elasticity of cloud facilitates resource quantity and characteristics to vary at runtime. It is associated

with the commissioning and decommissioning of resources. It is determined by the relative speed by

which resources are allocated (to some process) and are consumed.

What is the importance of multi-tenancy feature in any utility service?

Utility service model eliminates the need of maintaining separate and dedicated service generation

(as well as delivery) units for each and every consumer. In this model, everyone consume service from

some common source being offered by some service provider. Here, resources are used as demand

basis to serve consumers. A resource held by one consumer at a point of time may serve another

consumer when gets released. This approach makes the utility service cost-effective. Hence, multi-

tenancy is an integral part of utility service model.

Why multi-tenancy is not considered as an essential feature of cloud computing?

This is because of private cloud which does not have the scope for multi-tenancy of resources. In

public cloud arena, computing resources are usually shared with consumers. One resource held

by some consumer can be given to some other when it gets released from the first consumer. This

increases resource utilization and decreases cost of computing. But, in private cloud, all resources get

reserved for one consumer (or consumer organization), and not shared with others. This is why multi-

tenancy is not treated as an essential feature of cloud computing.

‘Private cloud is hosted within the security boundary of owner organization’ – Justify.

Enterprises who are concerned about the protection of their business interest often choose not

to share cloud environment with external organizations or external users. They work in private

cloud environment. Private cloud can be deployed on-premises or off-premises both. But the cloud (which is the resources of the cloud) even if it is hosted off-premises, cannot be shared with

anyone else apart from the owner enterprise. The cloud environment works as an integral part

within the security boundary of the owner organization to keep the computing environment more

protected.

Which one is more cost-effective: private cloud or public cloud?

Private cloud uses dedicated resources. Resources there cannot be shared with anyone else but the

owner (consumer). This guarantees a secured computing environment as storage and other resources

are not shared with others. But, this reservation of resources increases cost of computing. In public

cloud, resources are supplied to a consumer as per actual demand. No one can forcefully hold

resource without utilization and unutilized resources can be shared with any other consumers. This

multi-tenancy philosophy reduces cost of computing a lot.

How can organizations benefit from hybrid cloud?

Public cloud is cost-effective but private cloud can provide more security. Hence, many organizations

prefer to use private cloud. But, one problem often is being faced with private cloud during its peak

hours. The reserved resources may not meet the demand during such period of time. There we can

use hybrid cloud instead. In hybrid cloud, the enterprises can run the essential (or core) part of their

applications in the private cloud and expand into the public cloud utility when the demand goes

beyond a certain limit. Thus, they enjoy the security of private cloud by reducing the total cost.

Justify the 'low initial investment' philosophy of cloud computing.

In cloud computing, cloud vendors develop and keep all facilities ready for its potential customers

(the users). Vendors invest in large to develop the computing facilities those are essential to attract

customers. Thus, investment of cloud computing vendors is very high in its initial phase.

Users on the other hand can avail those facilities on rent basis as per the requirement. They need

not to procure resources or purchase software licenses any more. This reduces the initial investment

of users radically. Thus, the low investment philosophy of cloud is true from users' perspective.

Why is cloud computing called variable cost computing model?

Cloud computing philosophy enables users to use computing resources on rent basis. Unlike traditional computing, users need not to invest huge capital at the beginning for developing computing infrastructure and can pay to the cloud vendors as per their usage of resources. Resource

consumption is measured and billed. When use is higher, users pay accordingly and during low usage

they pay less. This is why cloud computing is called a variable cost computing model.

What are the advantages of utility computing?

Utility computing brings many benefits to users. It eliminates the burden of huge initial investment

from users. It allows them to pay as per their usage. Sharing of common resources among multiple

users under one computing infrastructure reduces cost of computing also. Users can simply connect and use computing. In brief, users get tension-free computing facilities at much lower

cost.

What is pervasive computing?

Pervasive computing refers to the computing environment having created through the use of embedded microprocessor that always remains connected with other devices via Internet. Devices

those are part of pervasive computing can be anything from car, electric meter to human body. Such

computing environment is generally built based on wireless communication technology, Internet

and advanced electronic chips. Pervasive computing devices always remain connected.

Is it true that cloud computing ensures round the clock availability?

It can be said that cloud computing is available anywhere and anytime. But it is true that cloud

computing vendors cannot generally deliver services with hundred percent service uptime assurance.

Reputed cloud vendors guarantee more than 99.9% of service uptime, but till now it has not been

possible for vendors to deliver 100% service uptime over a reasonable period of time.

Anyway, despite

of this, its service availability is still much better than the traditional computing scenario.

And cloud

users rarely suffer or can even feel the effect of the almost insignificant service downtime.

Why is cloud computing considered an evolution rather than as an innovation?

Cloud computing is the outcome of the convergence of research and developments in different

fields of computing. Advancements in distributed computing, emergence of resource virtualization

technique, developments in the field of web services and communication technology, arrival of SOA

paradigm in application development, advancements in the field of AI to automatically manage

computing infrastructure and related others, all of whom have contributed to realize the dream of cloud

computing. This is why cloud computing is considered as the next phase of evolution in computing.

How did the introduction of distributed system revolutionize computing?

Distributed systems introduced many features those are considered as revolutionary. It took the

parallel processing concept to next step where computing became faster. Faster processing results

in low latency. Formation of resource pool is another feature of distributed computing model. Apart

from this, distributed system opened the scope for making a system extra powerful by adding more

(virtually unlimited) resources (computing nodes) into the system. Thus, computing systems got its

scalable nature which turned a revolutionary feature with further developments.

Why is grid computing considered the predecessor of cloud computing?

Cloud computing is developed based on the distributed model of computing. There are several

methodologies those in cloud computing have been combined together to realize itself but

grid computing model was the first major breakthrough towards realizing the dream of cloud

computing system. This advanced mode of computing has inherited important characteristics like

creation of resource pool or system scalability from the grid computing model. This is why grid

computing is considered as the predecessor of cloud computing.

How application mobility has been possible with resource virtualization?

In traditional way of computing, once an application or process was launched, it used to get attached with

some physical resource(s) and fully or partially occupied the resource(s) till the completion or termination.

It was impossible to move a running application to a separate set of resources. Resource virtualization

decoupled application from physical computing resources. Application then did not have any direct

access or could not directly occupy hardware resources too. In between them there lies the virtualized

resources which can assign loads among physical resources. Thus, shifting load becomes possible.

Do enterprises need to purchase new hardware to shift into cloud?

For working in cloud, enterprises need low-configuration computing devices to be used as terminals.

The existing in-house machines (which are having high configurations) are more than sufficient to

be used as terminals for accessing cloud. It is likely that enterprises will have to make changes in

their networks while adopting cloud computing. Every terminals need to have high speed Internet

connectivity. This reorganization of the network may need some new procurement. But the organizations need not to purchase any significant volume of hardware while shifting in cloud.

What is meant by 'cloud' in cloud computing?

A cloud is combination of servers, storages, network devices (like switches etc.) networked together

from where computing facilities are delivered to users. Cloud facility is developed by activating this

specially designed network of commodity computing components with advanced software to power

them up for the task. This powered up facility becomes capable delivering computing as utility service

that user can access from any location through Internet.

Where do the clouds reside?

Globally cloud computing facilities are provided by major computing vendors. They generally develop

huge infrastructure for cloud at different locations around the globe. Such arrangements are known

as cloud data centers or CDCs. Apart from computing facilities, these centers have other essential

amenities like uninterrupted power supply system, fire control system and cooling system. CDCs are

highly protected areas where vendors set up networks of large number of computing devices like

servers, storages etc. and create cloud computing utility service out of those.

How can users gain from utility computing?

Utility computing is the way of supplying computing service through an on-demand, pay-per-usage

billing method. It focusses on the business model where users receive computing resources from

a service provider delivered over network communication channel. The prime advantage of utility

computing is its economic benefit that reduces users' computing cost.

Will cloud computing replace traditional computing devices with high-configured desktops or laptops?

In cloud computing approach, all load of computing shifts into the highly configured computing

facilities provided by the cloud. People just need an interfacing device through which they can

connect to the cloud using Internet. This interfacing device can be any desktop, laptop, tablet or

even a mobile interface which can run the cloud interfacing application. Hence, these interfacing

devices do not need to be highly configured any more. So, cloud computing is going to replace

highly-configured traditional desktops or laptops with minimally-configured ones, as they will only

have to act as interfacing devices to cloud.

Is cloud computing beneficial only for larger organizations?

Individual users who work in their PCs at home can avail cloud facility sitting at home itself. They

can store files, install software, or run programs just like they did it in PCs. Apart from individual

users, small offices that need a networked environment with a small number of terminals can easily

move into the cloud. In short, cloud computing is for all those who need computing facilities in any

form as well as volume.